

# Qatar-1b

R-2561

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_190629 · created 2026-08-26T23:11:20+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2458663.7624 +/- 0.0016 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.192 +/- 0.056                 |
| Transit depth (Rp/Rs)^2                           | 3.69 +/- 2.16 [%]               |
| Orbital Inclination (inc)                         | 88.83 +/- 2.84                  |
| Airmass coefficient 1 (a1)                        | 21.07 +/- 18.26                 |
| Airmass coefficient 2 (a2)                        | -2.18 +/- 0.51                  |
| Scatter in the residuals of the lightcurve fit is | 111.02 %                        |
| Best Comparison Star                              | #3 - [588, 369]                 |
| Optimal Aperture                                  | 0.99                            |
| Optimal Annulus                                   | 6.75                            |
| Transit Duration (day)                            | 0.0819 +/- 0.0062               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.192 $\pm$ 0.056 vs 0.1463 (deviation 0.29 $\sigma$ ) |
| Duration fit vs published            | 0.0819 d vs 0.0692 d (deviation 0.73 $\sigma$ )        |
| Mid-time O-C                         | -3.7 min                                               |
| Depth SNR                            | 1.2                                                    |
| Residual scatter                     | 111.02 %                                               |

### Light curve

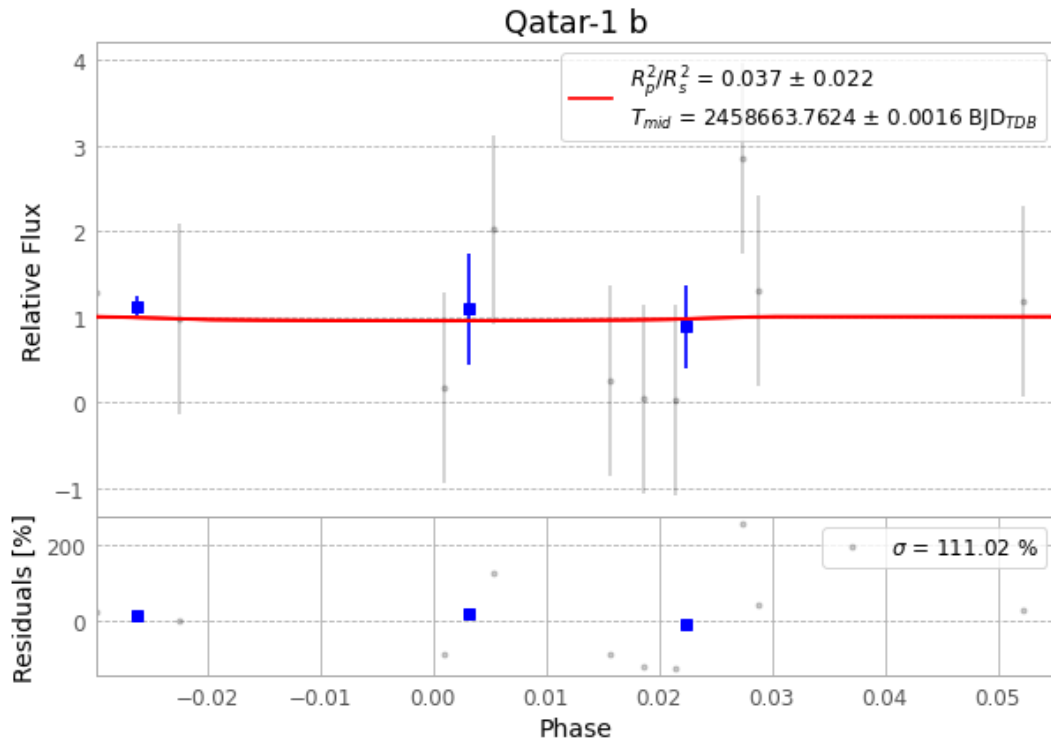


Figure 1 — FinalLightCurve\_Qatar-1 b\_2019-06-28.png (SHA-256 7f0e713ca33bbb33...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [563, 218], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 3cbbcd2e32536859...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

73 FITS frames (48 MB), Qatar-1190629043312.FITS → Qatar-1190629080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-190629.log (745ecf8c8afe...), FinalLightCurve\_Qatar-1 b\_2019-06-28.png (7f0e713ca33b...), FinalLightCurve\_Qatar-1 b\_2019-06-28.csv (9cae4abea2d7...)

## Compute resources

Wall time 205.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:11Z.